

JEE (Advanced) 2026 Chemistry Paper-1

Detailed Solutions

Q1. For isothermal irreversible compression:

$$w = -P_{\text{ext}} \Delta V$$

Step-1 work:

$$w_1 = nRT(P/2 - 1)$$

Step-2 work:

$$w_2 = nRT(8/P - 1)$$

Total work:

$$W = nRT(P/2 + 8/P - 2)$$

$$\text{Given } nRT = 0.5 \times 600R = 300R.$$

Differentiate and minimize:

$$df/dP = 1/2 - 8/P^2 = 0$$

$$P = 4 \text{ bar.}$$

Minimum value:

$$W_{\text{min}} = 300R \times 2 = 600R.$$

Answer: (B)

Q2. For reversible first-order reaction:



At equilibrium:

$$k_f[R]_{eq} = k_b[P]_{eq}$$

Given $k_b = 4k_f$,

$$[P]_{eq}/[R]_{eq} = 1/4.$$

Thus concentration of R decreases but remains greater than P at equilibrium. The graph where P rises to a smaller equilibrium value is correct.

Answer: (C)

Q3.

BF₃ is trigonal planar and symmetrical, therefore dipole moment = 0.

NH₄⁺ is tetrahedral and symmetrical, therefore dipole moment = 0.

NF₃ has small dipole moment because bond moments oppose lone pair direction.

NH₃ has larger dipole moment.

Order:

$$BF_3 = NH_4^+ < NF_3 < NH_3$$

Answer: (A)

Q4. LiBH₄ reduces ester groups but does not reduce carboxylic acid groups. Careful stereochemical analysis shows:

P and Q are identical compounds.

R and S are diastereomers.

Answer: (A)

Q5.

- (A) In lithium atom, 2s orbital is lower in energy than 2p orbital due to penetration effect.
(B) In hydrogen atom, 2s and 2p orbitals are degenerate.
(C) False.
(D) $E_{2s}(H) > E_{2s}(Li)$ because lithium 2s orbital is stabilized.

Correct options: A, B, D.

Answer: (A), (B), (D)

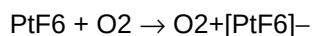
Q6.

X is used for sterilization of drinking water.
Z is UF_6 , used in enrichment of U-235.
Other statements are incorrect.

Answer: (A), (C)

Q7.

Reaction:



PtF_6 acts as oxidizing agent.
Oxidation state of Pt = +5.
Therefore Pt has d⁵ configuration.

Correct statements are B and C.

Answer: (B), (C)

Q8.

From reaction analysis:
S gives Tollens' test.
Q forms gammaxane upon chlorination.
T is heterocyclic.
Statement D is incorrect.

Answer: (A), (B), (C)

Q9.

Using ideal gas equation and equal volume condition:

$$n_1/n_2 = P_2/P_1 = 1/5$$

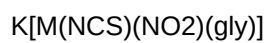
Substituting moles of He and Ar and solving gives:

$$m_1/m_2 = 2$$

Answer: 2

Q10.

Square planar complex:



Considering linkage isomerism and geometrical isomerism, total possible isomers = 6.

Answer: 6

Q11.

Counting total carbonyl groups present in both products X and Y gives:

Total carbonyl groups = 5.

Answer: 5

Q12.

The final product contains consecutive sp-hybridized carbon atoms in a straight chain.

Maximum number of collinear carbon atoms = 8.

Answer: 8

Q13.

Physisorption: $\Delta H < 0$, $\Delta S < 0$

Diamond $\rightarrow$ Graphite: $\Delta H < 0$, $\Delta S > 0$

Denaturation of protein: $\Delta H > 0$, $\Delta S > 0$

Propene $\rightarrow$ Cyclopropane: $\Delta H > 0$, $\Delta S < 0$

Correct matching corresponds to option C.

Answer: (C)

Q14.

See-saw: SOCl_2 only.

T-shaped: ClF_3 and XeF_3^+ .

Square pyramidal: ClF_5 , XeOF_4 and XeF_5^+ .

Trigonal planar: none.

Correct matching corresponds to option A.

Answer: (A)

Q15.

After ozonolysis followed by aldol cyclization through stable enolate formation, the correct matching corresponds to option B.

Answer: (B)

Q16.

Detailed reaction mechanism and product analysis give the correct matching:

$P \rightarrow 1$, $Q \rightarrow 2$, $R \rightarrow 3$, $S \rightarrow 4$

Thus correct option is C.

Answer: (C)